Partitioning $\{1, \ldots, n\}$ into parts of size at most three whose sums are powers of three

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Abstract

We prove the conjecture of Gurvich and Naumova that for every $n \geq 1$ the set $\{1, \ldots, n\}$ has a partition into parts of at most three elements whose sums are powers of 3. The partition is given by an explicit recursion. The proof is formalized in Lean 4.

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1. Introduction

Let $m \geq 2$ and $n \geq 1$ be integers and $[n] = \{1, \ldots, n\}$. Gurvich and Naumova [1] call a partition of $[n]$ *m-good* if every part has at most m elements and the sum of the elements of every part is a power m^k, $k \geq 0$, of m. Every $[n]$ has exactly one 2-good partition [1, Proposition 1]. In an m-good partition every part other than $\{1\}$ has sum $\equiv 0 \pmod{m}$, so $n(n+1)/2 \equiv 0$ or $1 \pmod{m}$. For $n = 2(m^t + 1)$ this sum is $\equiv 3 \pmod{m}$; hence for $m \geq 4$ the set $[n]$ has no m-good partition [1, Observation 1]. For $m = 3$ they stated:

Conjecture 1.1 ([1, Conjecture 1])**.** *For every $n \geq 1$ the set $[n]$ has a 3-good partition.*

They proved that a minimal counterexample is of the form $n = 3^t + 3k + 2$ with $0 \leq k < (3^{t-1} - 1)/2$ and $k \neq (3^\ell - 1)/2$ for $0 \leq \ell < t$ [1, Theorem 1],

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recorded further reductions found by O. Polubasoff, and reported that a computation by D. Rybin confirmed the conjecture for $n \le 844$. We prove it.

Theorem 1.2. *For every $n \ge 1$ the set $[n]$ has a 3-good partition.*

Corollary 1.3. *Let $m \ge 2$. Every set $[n]$, $n \ge 1$, has an m-good partition if and only if $m \in \{2, 3\}$.*

The proof is a recursion on n. Let $T = 3^t \le n < 3T$. For $n = T$, for $2n > 3T$, and for $n = T + j$ with $j \not\equiv 2 \pmod 3$, the reductions of [1, Section 2] pass from $[n]$ to a shorter interval. For the remaining $n = T + j$, $j \equiv 2 \pmod 3$, $2j < T$, we prove more: $[n]$ minus the triple $\{n-1, n, T-2j+1\}$, whose sum is $3T$, has a 3-good partition (Proposition 4.1). This statement passes from j to n by a reflection argument (Lemma 3.1), except when j lies in the upper half of the interval $[3^u, 3^{u+1})$ that contains it. In that case a fixed-point-free involution of $\{1, \dots, 2N-2\}$ with distinct pair sums supplies the missing triples (Lemma 3.2). Section 5 describes the Lean 4 formalization and the computations.

2. Good and balanced sets

For integers $a \le b$ let $[a, b] = \{x \in \mathbb{Z} : a \le x \le b\}$, and $[a, b] = \emptyset$ if $a > b$. For $Z \subset \mathbb{Z}$ and $c, s \in \mathbb{Z}$ write $c + sZ = \{c + sz : z \in Z\}$ and $c - Z = \{c - z : z \in Z\}$. A finite set $A \subset \mathbb{Z}$ is *admissible* if $|A| \le 3$ and $\sum_{x \in A} x = 3^k$ for some $k \ge 0$. A finite set $S \subset \mathbb{Z}$ is *good* if it is a disjoint union of admissible sets; thus $[n]$ is good if and only if it has a 3-good partition. A finite set $Z \subset \mathbb{Z}$ is *balanced* if it is a disjoint union of sets each of which is $\{0\}$ or a 3-element set with sum 0. Disjoint unions of good sets are good, and disjoint unions of balanced sets are balanced.

Lemma 2.1 ([1, Lemma 1])**.** *If $\ell \ge 0$ and $\ell \not\equiv 2 \pmod 3$, then $[-\ell, \ell]$ is balanced.*

Proof. Write $\ell = 3k + r$ with $r \in \{0, 1\}$. For $0 \le i < k$ the sets

$$A_i = \{-(3k-i),\ 1+i,\ 3k-1-2i\}, \qquad B_i = \{-(2k-i),\ -(k-i),\ 3k-2i\}$$

have sum 0. As i runs over $[0, k-1]$, the entries $-(3k-i)$, $-(2k-i)$, $-(k-i)$ run over $[-3k, -2k-1]$, $[-2k, -k-1]$, $[-k, -1]$, the entries $1+i$

run over $[1,k]$, and the entries $3k-1-2i$ and $3k-2i$ run over the numbers in $[k+1,3k]$ of the parity of $3k-1$ and of $3k$. So the $2k$ sets A_i, B_i partition $[-3k,3k]\setminus\{0\}$. Add $\{0\}$ if $r=0$ and $\{-\ell,0,\ell\}$ if $r=1$. □

In the proof in [1] the two sums over i start at $i=1$; they start at $i=0$, as the examples $k=1,2$ given there show.

Lemma 2.2. *Let Z be balanced, $q\ge 0$ and $s\neq 0$. Then 3^q+sZ is good and sZ is balanced.*

Proof. The map $x\mapsto 3^q+sx$ is injective; it sends $\{0\}$ to $\{3^q\}$ and a 3-element set with sum 0 to a 3-element set with sum 3^{q+1}. The map $x\mapsto sx$ is injective and preserves the two kinds of parts. □

Lemma 2.3. *Let $Q=3^q$ and let $X\subseteq[1,Q-1]$ satisfy $Q-X=X$. Then X is good.*

Proof. Since Q is odd, $x\neq Q-x$, and X is the disjoint union of the pairs $\{x,Q-x\}$ with $x\in X$, $x<Q/2$; each has sum Q. □

3. Reflection and the involutions σ_N

Lemma 3.1. *Let $T=3^t$, $j\ge 0$, $D\subseteq[1,T-1]$ and $W\subseteq[T+1,T+j]\cup(T-D)$. Put*

$$F=\{d\in D: T-d\notin D\},\qquad R=\big(\{T\}\cup[T+1,T+j]\cup(T-D)\big)\setminus W.$$

If F and R are good, then $[1,T+j]\setminus W$ is good.

Proof. Let $E=\{x\in[1,T-1]: x\notin D,\ T-x\notin D\}$. Then $T-E=E$, so E is good by Lemma 2.3. We show that $[1,T+j]\setminus W$ is the disjoint union of E, F and R. The set E meets neither $D\supseteq F$ nor $\{T\}\cup[T+1,T+j]\cup(T-D)\supseteq R$; and $F\subseteq[1,T-1]$ does not meet $T-D$, hence meets neither R nor W. Since E does not meet W either, E, F, R are disjoint subsets of $[1,T+j]\setminus W$. Conversely let $x\in[1,T+j]\setminus W$. If $x\ge T$, or if $x\le T-1$ and $T-x\in D$, then $x\in R$. Otherwise $x\le T-1$, $T-x\notin D$, and $x\in F$ or $x\in E$ according as $x\in D$ or not. □

Lemma 3.2. *Let $N\ge 1$, $N\equiv 1 \pmod 3$, and $K=2N-2$. There is a fixed-point-free involution σ of $[1,K]$ with the following properties, where $\Pi=\{z\in[1,K]: z<\sigma(z)\}$.*

(i) *The sums* $z+\sigma(z)$, $z\in\Pi$, *are distinct.*

(ii) *The* N *numbers* $2N-2$ *and* $z+\sigma(z)-(2N+1)$, $z\in\Pi$, *are distinct, and the set* Δ_N *they form is balanced.*

(iii) *Every* $\delta\in\Delta_N$ *satisfies* $|\delta|\le 2N-2$, *and every positive* $\delta\in\Delta_N$ *is even.*

Proof. Since $N\equiv 1 \pmod 3$, either $N=1$, or $N=2M$ with $M\ge 2$ and $M\equiv 2 \pmod 3$, or $N=2M+1$ with $M\ge 3$ and $M\equiv 0 \pmod 3$.

If $N=1$, then $[1,K]=\emptyset$ and $\Delta_1=\{0\}$.

Let $N=2M$. Let σ exchange $1\leftrightarrow N$, $2\leftrightarrow N+1$, and $x\leftrightarrow x+N-1$ for $3\le x\le N-1$. These pairs partition $[1,K]$: the last family pairs $[3,N-1]$ with $[N+2,2N-2]$. The pair sums are $N+1$, $N+3$ and $2x+N-1$, $3\le x\le N-1$, which are distinct. Subtracting $2N+1=4M+1$ gives

$$\Delta_N=\{4M-2,\,-2M,\,-(2M-2)\}\cup 2\,[-(M-2),M-2].$$

The first set has sum 0; the second lies in $[-(2M-4),2M-4]$, so it does not meet the first, and it is balanced by Lemmas 2.1 and 2.2, since $M-2\equiv 0 \pmod 3$. This proves (i)–(iii), as $4M-2=2N-2$.

Let $N=2M+1$. Let σ exchange $1\leftrightarrow N$, $2\leftrightarrow N+1$, $3\leftrightarrow N-1$, $4\leftrightarrow N+2$, $2N-3\leftrightarrow 2N-2$, and $x\leftrightarrow x+N-2$ for $5\le x\le N-2$. The last family pairs $[5,N-2]$ with $[N+3,2N-4]$, so these pairs partition $[1,K]$. The pair sums are $N+1$, $N+3$, $N+2$, $N+6$, $4N-5$ and $2x+N-2\in[N+8,3N-6]$, $5\le x\le N-2$, which are distinct. Subtracting $2N+1=4M+3$ gives

$$\Delta_N=\{4M,\,-(2M+1),\,-(2M-1)\}\cup\{4M-4,\,-2M,\,-(2M-4)\}\\ \cup 2\,[-(M-3),M-3].$$

The first two sets have sum 0; the third lies in $[-(2M-6),2M-6]$ and is balanced since $M-3\equiv 0 \pmod 3$. The three sets are disjoint, which proves (i)–(iii), as $4M=2N-2$. □

4. Proof of Theorem 1.2

For $n\ge 1$ let t be the integer with $3^t\le n<3^{t+1}$. If $t\ge 1$ and $j=n-3^t$ satisfies

$$j\equiv 2 \pmod 3,\qquad 2\le j,\qquad 2j+1\le 3^t, \tag{1}$$

put $W_n=\{n-1,\,n,\,3^t-2j+1\}$. Its elements are distinct, since $2\le 3^t-2j+1\le 3^t-3$, and their sum is 3^{t+1}. Let $H(n)$ be the statement that $[n]\setminus W_n$ is good.

Proposition 4.1. *Let $n = 3^t + j$ with $t \geq 1$ and j as in (1). Suppose that $[m]$ is good for every $m < n$ and that $H(m)$ holds for every $m < n$ for which it is defined. Then $H(n)$ holds.*

Proof. Let $T = 3^t$, so $W_n = \{T+j-1, T+j, T-2j+1\}$. Since $T \geq 2j+1 \geq 5$, we have $T \geq 9$. In each case below we choose $D \subseteq [1, T-1]$ with $2j-1 \in D$, so that $W = W_n \subseteq [T+1, T+j] \cup (T-D)$, and show that the sets F and R of Lemma 3.1 are good. Lemma 3.1 then gives $H(n)$.

Case $j = 2$. Take $D = \{3\}$. Then $F = \{3\}$, as $T - 3 \neq 3$, and $R = \{T, T+1, T+2, T-3\} \setminus W = \{T\}$.

For $j \geq 5$ let u be the integer with $3^u \leq j < 3^{u+1}$. Then $u \geq 1$, and $j \geq 3^u + 2$ because $j \equiv 2$ and $3^u \equiv 0 \pmod 3$. Also $3j \neq T$, since $3j = T$ would give $j = 3^{t-1} \equiv 0 \pmod 3$.

Cases $3j > T$, and $3j < T$ with $2j + 1 \leq 3^{u+1}$. Take $D = [1, j-2] \cup \{2j-1\}$. Then $T - D = [T-j+2, T-1] \cup \{T-2j+1\}$ and

$$R = T + [-(j-2), j-2],$$

which is good by Lemmas 2.1 and 2.2, since $j - 2 \equiv 0 \pmod 3$.

If $3j > T$, then $u = t - 1$. As T and $3j$ are multiples of 3, $T \leq 3j - 3$, so $T-2j+1 \in [2, j-2] \subseteq D$. For $d \in [1, j-2]$ we have $T-d \geq T-j+2 > j-2$, so $T - d \in D$ only if $T - d = 2j - 1$. Hence

$$F = [1, j-2] \setminus \{T - 2j + 1\} = [j] \setminus W_j,$$

because $3^{u+1} = T$ and $W_j = \{j-1, j, T-2j+1\}$. The number $j = 3^u + (j - 3^u)$ satisfies (1) at level u, so F is good by $H(j)$.

If $3j < T$ and $2j + 1 \leq 3^{u+1}$, then $3^{u+1} \leq 3j < T$ gives $T \geq 3^{u+2}$, and every $d \in D$ satisfies $T - d \geq T - 2j + 1 > 3^{u+2} - 3^{u+1} > 2j - 1$. Hence $F = D$. Put $p = 3^{u+1} - 2j + 1$; then $2 \leq p \leq j - 2$, because $2j + 1 \leq 3^{u+1}$ and $j \geq 3^u + 1$. Since $W_j = \{j-1, j, p\}$,

$$F = D = \big([j] \setminus W_j\big) \cup \{p,\, 2j-1\},$$

a disjoint union of a set that is good by $H(j)$ and a pair with sum 3^{u+1}.

Case $3j < T$ and $2j + 1 > 3^{u+1}$. Let $P = 3^{u+1}$. As before $T \geq 3P$, so $T = 3P$ or $T \geq 9P$; and $P + 1 \leq 2j \leq 2P - 2$. Write $P = 2c + 1$ and put

$$N = j - c, \qquad e = P - 1 - j = c - N.$$

Then $N \geq 1$, $e \geq 0$, $j = e + 2N$, and $2N - 2 = 2j - P - 1 \leq P - 3$. Since $3 \mid P$ we have $c \equiv 1 \pmod 3$, hence $N \equiv 1$ and $e \equiv 0 \pmod 3$. Let σ, Π, Δ_N be as in Lemma 3.2, $K = 2N - 2$, and

$$D = [1, e] \cup (P + \Delta_N).$$

By Lemma 3.2(iii), $P + \Delta_N \subseteq [P - 2N + 2, P + 2N - 2] = [2e + 3, 2j - 1]$, so the union is disjoint, $D \subseteq [1, 2P - 3] \subseteq [1, T - 1]$, and $2j - 1 = P + (2N - 2) \in D$.

We show $F = D$. Let $d, d' \in D$ with $d + d' = T$. If $d \leq e$, then $d + d' \leq e + 2P - 3 < 3P \leq T$; so $d = P + \delta$ and $d' = P + \delta'$ with $\delta, \delta' \in \Delta_N$ and $\delta + \delta' = T - 2P$. If $T \geq 9P$, then $\delta + \delta' \geq 7P > 2(P - 3)$, which is impossible. If $T = 3P$, then $\delta + \delta' = P$ is odd. Since $\delta, \delta' \leq P - 3$, both are positive, hence even by Lemma 3.2(iii), a contradiction. Thus $F = D$, which is good: $[1, e] = [e]$ is good by hypothesis, and $P + \Delta_N$ by Lemmas 3.2 and 2.2.

It remains to compute R. First, $[T + 1, T + j] \setminus W = T + [1, j - 2] = T + [1, e + K]$. Second, $T - D = T + [-e, -1]$ together with $T - P - \Delta_N$; here $T - P - (2N - 2) = T - 2j + 1 \in W$, and for $z \in \Pi$, using $P - 2N - 1 = 2e$,

$$T - P - \big(z + \sigma(z) - (2N + 1)\big) = T - \big(2e + z + \sigma(z)\big).$$

By Lemma 3.2(ii) these numbers differ from $T - 2j + 1$. Hence $R = T + Z$ with

$$Z = [-e, e] \ \cup \bigcup_{z \in \Pi} \big\{e + z,\ e + \sigma(z),\ -(2e + z + \sigma(z))\big\}.$$

Each set $\{e + z, e + \sigma(z), -(2e + z + \sigma(z))\}$ has three elements and sum 0. As z runs over Π, the numbers $e + z$ and $e + \sigma(z)$ run over $e + [1, K]$ once each, because σ is a fixed-point-free involution; the numbers $-(2e + z + \sigma(z))$ are distinct by Lemma 3.2(i) and less than $-e$. Since $[-e, e]$ is balanced by Lemma 2.1, Z is balanced, and R is good by Lemma 2.2. □

Proof of Theorem 1.2. We prove by strong induction on $n \geq 0$ that $[n]$ is good and that $H(n)$ holds whenever it is defined; $[0] = \emptyset$. The sets $[1] = \{1\}$ and $[2] = \{1, 2\}$ are admissible, and $H(1)$, $H(2)$ are not defined. Let $n \geq 3$, $T = 3^t \leq n < 3T$, $t \geq 1$. Whenever $H(n)$ is defined, it holds by Proposition 4.1 and the induction hypothesis. For $[n]$ there are four cases.

(a) $n = T$: $[n] = [T - 1] \cup \{T\}$.

(b) $2n \geq 3T + 1$: with $a = 3T - 1 - n < n$, $[n] = [a] \cup [a + 1, 3T - 1 - a]$, and the second set is good by Lemma 2.3 with $Q = 3T$.

(c) $T < n$, $2n \le 3T$ and $j = n - T \not\equiv 2 \pmod 3$: $[n]$ is the union of $[T - j - 1]$ and $T + [-j, j]$, and the second set is good by Lemmas 2.1 and 2.2.

(d) $T < n$, $2n \le 3T$ and $j = n - T \equiv 2 \pmod 3$: then j satisfies (1), and $[n] = ([n] \setminus W_n) \cup W_n$, where $[n] \setminus W_n$ is good by $H(n)$ and W_n is admissible.

In (a)–(c) the first set is good by the induction hypothesis, and in all cases the union is disjoint. □

Proof of Corollary 1.3. For $m = 2$ this is [1, Proposition 1], for $m = 3$ it is Theorem 1.2, and for $m \ge 4$ the set $[2m + 2]$ has no m-good partition, as recalled in Section 1. □

Example 4.2. The least n not excluded by [1, Theorem 1] is $n = 35 = 27{+}8$. Here $3j = 24 < 27$, $u = 1$ and $2j + 1 = 17 > 9$, so the last case applies with $P = 9$, $c = 4$, $N = 4$, $e = 0$. The involution σ of $[1, 6]$ is $1 \leftrightarrow 4$, $2 \leftrightarrow 5$, $3 \leftrightarrow 6$, $\Delta_4 = \{6, -4, -2, 0\}$ and $D = \{5, 7, 9, 15\}$. The construction gives the 3-good partition of [35] into

$$\{5, 7, 15\},\ \{9\},\ \{27\},\ \{18, 30, 33\},\ \{20, 29, 32\},\ \{22, 28, 31\},\ \{12, 34, 35\}$$

and the nine pairs $\{x, 27 - x\}$, $x \in \{1, 2, 3, 4, 6, 8, 10, 11, 13\}$.

The recursion computes the partition of $[n]$ in $O(n)$ arithmetic operations.

5. Formal verification and computations

Theorem 1.2 is formalized in Lean 4 [2] with Mathlib [3] (Lean and Mathlib version 4.34.1). The source, seven files with 1073 lines, is in the directory `lean` of the archive described under Data availability. Sets are finite sets of integers; good and balanced sets are inductive predicates generated from the empty set and the admissible sets, respectively the sets $\{0\}$ and 3-element sets with sum 0, by disjoint union. The file `Basic.lean` contains Lemmas 2.1–2.3, `Gadget.lean` Lemma 3.2, `Assembly.lean` Lemma 3.1 with the first two cases of Proposition 4.1, `CaseA2.lean` its last case, and `Main.lean` the induction of Section 4 and the main theorem:

```
theorem gurvich_naumova (n : ℕ) :
    ∃ P : Finpartition (Finset.Icc 1 n),
      ∀ s ∈ P.parts, s.card ≤ 3 ∧ ∃ k : ℕ, ∑ x ∈ s, x = 3 ^ k
```

Here `Finpartition (Finset.Icc 1 n)` is the type of partitions of $[n]$ into nonempty parts. The command `lake build` compiles the project and runs

```
#print axioms GN.gurvich_naumova
```

which reports `propext`, `Classical.choice` and `Quot.sound`, the axioms of Lean's logic. The source contains no `sorry` and declares no axiom.

Programs in Python, C, C++ and Julia, included in the archive, implement the construction of Section 4. They check that it gives a 3-good partition of $[n]$ for every $n \le 30000$ (C; 20000 in Julia, 10000 in C++, 3000 in Python) and for larger values of n up to $6.5 \cdot 10^7$, check $H(n)$ for every n in these ranges for which it is defined, and check Lemma 2.1 for $\ell \le 3000$ and Lemma 3.2 for $N \le 3001$ (Python; $\ell \le 30000$ and $N \le 30001$ in C++). A shell script runs the Lean build, the axiom check and all programs. The construction, the proofs, the formalization and the programs were developed with Claude Code (Anthropic); see the declaration below.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Code (Anthropic) in order to develop the construction and the proof, write the Lean formalization and the verification programs, and draft the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Data availability

The repository `https://github.com/creelie/didactic-lamp` contains the Lean 4 formalization, the verification programs and the shell script that runs them. It is archived on Zenodo under doi:10.5281/zenodo.23316376 (all versions; release v1.0.0 is doi:10.5281/zenodo.23316377).

References

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